

Borla

Technical Debt Plan

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1. How debt was tracked during the build

Every deliberate shortcut was written down **as the decision was made**, not reconstructed afterwards — most of the items below are quoted almost verbatim from code comments left at the exact line where the trade-off was taken (e.g. `server/src/jobs/index.ts`, `server/src/modules/auth/routes.ts`). This is the practice the course asks for: technical debt as a first-class, visible decision, not a silent shortcut discovered later.

Each item below follows **Debt → Cause → Impact → Priority → Proposed Resolution**, and is labelled as one of:

- **Acceptable temporarily** — correct trade-off for a graded demo at pilot scale; revisit only if the project continues past this exam.
- **Scheduled for future resolution** — fine for now, but must be fixed before real users or a meaningful scale-up.
- **Critical** — must not reach real users in its current form.

2. Debt register

ID	Debt	Cause	Impact	Priority	Proposed resolution
TD-01	Review/reply moderation runs with no AI key by default — everything lands in a manual admin queue instead of being auto-screened.	No Google AI Studio account was provisioned for this build; the moderation pipeline (<code>server/src/ai/moderation.ts</code>) checks <code>GEMINI_API_KEY</code> and no-ops if absent.	Every review needs a human click before publishing; fine at demo volume (a handful of reviews), would not scale to real usage — but is explicitly the <i>safer</i> failure direction (fail-closed, not fail-open).	● Scheduled	Provision a Gemini API key on Render (<code>GEMINI_API_KEY</code> env var, <code>sync:false</code> in <code>render.yaml</code> so it must be set manually) — the code path is already written and tested; this is a config change, not a code change.
TD-02	No real SMS gateway — mostly resolved: <code>server/src/utisls/sms.ts</code> sends the OTP via GiantSMS when <code>GIANTSMS_API_TOKEN</code> / <code>GIANTSMS_SENDER_ID</code> are configured, falling back to the in-app <code>devOtp</code> display only if the send fails or isn't configured. Confirmed live in production: <code>POST /api/auth/otp/request</code> against <code>https://borla.onrender.com</code> returned <code>{ "delivered": true }</code> — GiantSMS accepted the reconstructed request shape (auth header, <code>from / to / msg body</code>) for real, not just in theory. A related gap closed at the same time: the two seeded demo numbers are arbitrary, not real handsets, so once real SMS was wired up they would have started reporting <code>delivered:true</code> while the code vanished into a gateway with no phone behind it — a graded demo would have been locked out. Fixed by special-casing <code>DEMO_PHONES</code> (<code>server/src/seed.ts</code>) to always use a fixed, documented code (<code>DEMO_OTP</code>) and never attempt a real send, regardless of gateway state.	The HTTP contract was reconstructed from published third-party client libraries, not a first-party OpenAPI spec. Production accepted the request; actual handset delivery to a <i>real</i> number has still not been visually confirmed (by definition, the two numbers this build can freely test against are the exempted demo ones).	Low residual risk: the gateway accepting the request end-to-end (not just returning a generic 200) is strong evidence the integration is correct for a genuine phone number, but "accepted by the gateway" and "arrived on a phone" are not identical.	● Scheduled (down from Critical — a real send now provably works at the HTTP layer, and grading no longer depends on it working at all)	Confirm one delivery to a real handset, then remove the <code>devOtp</code> fallback field for non-demo numbers entirely so a failed send surfaces as a retryable error instead of a security bypass.
TD-03	Native background geolocation was not built. Collectors only report position while their browser tab is open and in the foreground (<code>useGeolocation</code> <code>watchPosition</code> , no Capacitor/native service).	Requires an Android toolchain, a background-location plugin (commercial or complex to configure), and a physical/emulated device to validate "screen-off roaming" — none available, and a native APK isn't gradable as a "Live Application URL" per the exam's own submission format.	This is the single biggest gap from the original design's own risk register (<code>borla-technical-design.md</code> §14, risk #1) — a collector who backgrounds the app or locks their phone stops updating position, which is exactly the failure mode that erodes trust in the map.	Critical (for any real pilot) / ● Acceptable for this exam	Wrap the existing shared React screens in Capacitor, add <code>@capacitor-community/background-geolocation</code> , and test on real low-end Android hardware (Tecno/Infinix/itel) per the original risk mitigation — no server-side change needed, since the API already accepts position updates from any authenticated client.
TD-04	No offline PWA shell — mostly resolved: <code>vite-plugin-pwa</code> now ships a real installable PWA — manifest + icons (<code>client/public/icon-*.png</code>), a Workbox service worker precaching the app shell (12 entries, ~490KB), an explicit "Update available" banner (<code>registerType: 'prompt'</code> , <code>src/pwa/UpdatePrompt.tsx</code> — never silently swaps content under an open tab) and a real "Install app" button (<code>src/pwa/InstallButton.tsx</code> , <code>beforeinstallprompt</code>). Verified: the service worker registers and activates (checked via a headless-browser <code>navigator.serviceWorker.getRegistrations()</code> call against the built <code>dist/</code>), and API/socket traffic is deliberately excluded from precaching so it's never served stale. Still open: households' geolocation is on-demand/foreground-only (unchanged), and there's no queued-writes-while-offline behaviour yet (e.g. "clear pin" typed while	The remaining gap is genuinely more work (background sync / an offline action queue), not a missing dependency.	Households still see a blank state for anything requiring a live fetch while offline (map data, requests, broadcasts) — the shell now loads, the <i>data</i> doesn't.	● Acceptable (down from Scheduled — the install/update half is done; only the offline-writes half remains)	Add a Workbox Background Sync queue (or a hand-rolled IndexedDB outbox) for "clear pin" and similarly small mutating actions so they survive a dropped connection.

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	offline doesn't queue and flush on reconnect — the shell loads offline, but mutating actions still need a live connection).				
TD-05	No Redis/BullMQ. Presence TTL, pin expiry, request timeout, and double-blind review release are all <code>node-cron</code> sweeps against Postgres directly; rate-limiting is a coarse in-memory per-IP limiter (<code>express-rate-limit</code>), not the design's per-collector notification cap.	Provisioning a managed Redis instance is an extra account/service with no functional benefit at this traffic volume; BullMQ adds retry/backoff semantics that matter only once notification volume is real.	All scheduled-job state is lost on process restart (a mid-flight sweep just re-runs on the next tick — acceptable, since every sweep is idempotent by construction) and cannot scale past one Node instance (Socket.IO has no Redis adapter either, so multi-instance deployments would silently drop events to users connected to a different instance).	● Scheduled	Add Upstash/Render Redis, move the four cron sweeps to BullMQ repeatable/delayed jobs, wire the Socket.IO Redis adapter, and replace the per-IP limiter with the design's per-collector <code>ratelimit:notif:{id}</code> counter — all four sweeps already have clean, small, idempotent bodies (<code>server/src/jobs/index.ts</code>) so the migration is mostly infrastructure, not logic rewrite.
TD-06	Admin auth has no 2FA — phone + password only (<code>bcrypt</code> -hashed), same JWT mechanism as household/collector, instead of the original design's "stronger auth than the field apps."	Full 2FA (TOTP/SMS) needs either the same missing SMS gateway (TD-02) or a new TOTP flow — both extra scope for an account class that, in this build, is a single seeded demo user.	An admin account compromise is higher blast radius than a household/collector one (can suspend users, remove content) — acceptable only because there is exactly one demo admin account behind a strong generated password, never exposed publicly except in the graded credentials file.	● Critical (before real ops staff use it)	Add TOTP (e.g. <code>otplib</code>) gated behind the existing <code>role='admin'</code> check; store a <code>totp_secret</code> column already anticipated by the schema's extensibility (JSONB <code>app_config</code> pattern could hold it, or a dedicated column).
TD-07	The standalone <code>admins</code> table from the original design was folded into <code>users.role = 'admin'</code> with a nullable <code>password_hash</code> column that only admins use.	Reduces schema surface for a single-admin demo; avoids a parallel user model.	Slight modelling smell — <code>password_hash</code> is <code>NULL</code> for 99% of rows (households/collectors) — and there is no dedicated <code>admins.role_tier</code> (<code>ops / moderator / superadmin</code> from the original design).	● Acceptable	If multiple admin tiers are ever needed, split back into a dedicated <code>admins</code> table with its own role enum, migrated via a straightforward <code>INSERT ... SELECT</code> from <code>users</code> WHERE <code>role='admin'</code> .
TD-08	Leaflet + raw OpenStreetMap raster tiles instead of the original design's MapLibre GL + vector tiles.	Zero-config; no tile-provider account/key needed, which matters when nobody is provisioning new accounts mid-exam.	Slightly less crisp rendering and no offline vector caching; functionally equivalent for markers/popups/radius visualisation, which is all this build needs.	● Acceptable	Swap the <code>MapView</code> component's tile layer for a MapLibre + vector source once a tile provider (e.g. <code>MapTiler</code> free tier) is chosen — the component's public interface (<code>center</code> , <code>points</code>) would not need to change.
TD-09	No i18n beyond English , despite the original design and the low-literacy design brief calling for Twi/Ga from day one.	Time-boxed; English-only was the fastest path to a fully functional, testable app across every screen.	Directly works against the stated accessibility goal for the target users.	● Scheduled	The UI already isolates copy inside JSX rather than scattering it — introduce <code>react-i18next</code> (or the originally-planned <code>packages/shared/i18n</code>) and extract strings; start with Twi and Ga per the pilot-neighbourhood plan.
TD-10	Automated test coverage is broad, not deep. 31 server tests cover every route's happy path and its most important failure mode (idempotency guards, role gates, moderation gating); there is no fuzzing, no load testing, and client tests cover 2 of ~9 components.	A time-boxed exam favoured "every module has real integration tests" over "one module is exhaustively tested" — a deliberate breadth-over-depth call, consistent with §7.7 of the SRS.	Edge cases in less-tested paths (e.g. concurrent double-accept races under real network latency, malformed geo payloads) are not proven safe, only reasoned about.	● Scheduled	Add property-based tests for the geospatial radius math, a concurrency test that fires <code>accept+reject</code> simultaneously against the same request to prove the conditional-update guard under real contention (not just sequential calls), and React Testing Library coverage for <code>HouseholdHome / CollectorHome</code> .
TD-11	Photo → waste-type AI classification was not built (§18 Job B in the original design).	Explicitly deprioritised behind moderation (Job A) per the original design's own value ranking — moderation is the safety-critical one; photo classification is a UX nicety.	Households still pick a waste-type tile manually instead of snapping a photo — a real but non-blocking gap for low-literacy accessibility.	● Scheduled	Reuse the existing <code>aiClient</code> -shaped moderation module's pattern: a second thin wrapper calling Gemini's multimodal endpoint, wired to a new <code>POST /broadcasts</code> image field, client-side downscaled before upload.
TD-12	Provider call-masking was not built — accepted requests reveal each party's real phone number ("reveal-on-accept", the original design's own recommended v1 approach, §10 option 1).	This is the design's own recommended v1 trade-off, not a shortcut taken under exam pressure — kept as-is deliberately.	Real phone numbers are exchanged between two people who chose each other via an accepted request — acceptable given the low stakes described in the original design.	● Acceptable	Add provider number-masking (e.g. via Hubtel) only if abuse reports or scale warrant the added per-call cost, exactly as the original design already prescribes.
TD-13	Single Render Web Service, no staging environment, no CI pipeline beyond local <code>npm test</code> .	Free-tier, single-service deploy was the fastest path to a real "Live	A bad commit reaches production directly; no automated gate before deploy.	● Scheduled	Add a GitHub Actions workflow running <code>npm test</code> on every push (config-only change, no app-code

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		Application URL" within the exam window.			change needed) and a second Render environment for staging before promoting to the graded URL.

3. Debt NOT taken — deliberate design choices worth naming

To be explicit about the difference between *debt* (a shortcut that should eventually be paid down) and a *considered trade-off that stays*: the no-collector-binding broadcast plane, the absence of in-app payments, and reveal-on-accept-instead-of-provider-masking are not technical debt — they are the original design's own architecture, kept unchanged because they are still the right call at this scale (see `borla-technical-design.md` §1 and §10, and TD-12 above).

4. Repayment plan by phase

This maps directly onto `Project_Documentation.md` §15 (Maintenance & Future Evolution):

Phase	Items paid down
v1.1 (immediate post-submission, if continued)	TD-06 (admin 2FA) — the one remaining 🔴 Critical item, must land before any real admin touches the system; confirm one real GiantSMS handset delivery to fully close TD-02
v1.2	TD-01 (provision Gemini key in production), TD-13 (CI pipeline)
v1.3	TD-03 (native Capacitor collector app + background geolocation) — the single biggest lift, matched to the original design's own phased roadmap
v2.0	TD-05 (Redis/BullMQ + Socket.IO Redis adapter, needed once traffic outgrows one instance), TD-09 (i18n), TD-04 (offline write queue — the install/update half already shipped), TD-11 (photo classification)
Ongoing	TD-10 (expand test depth every time a new bug class is found in production)